

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1022+850

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MNB

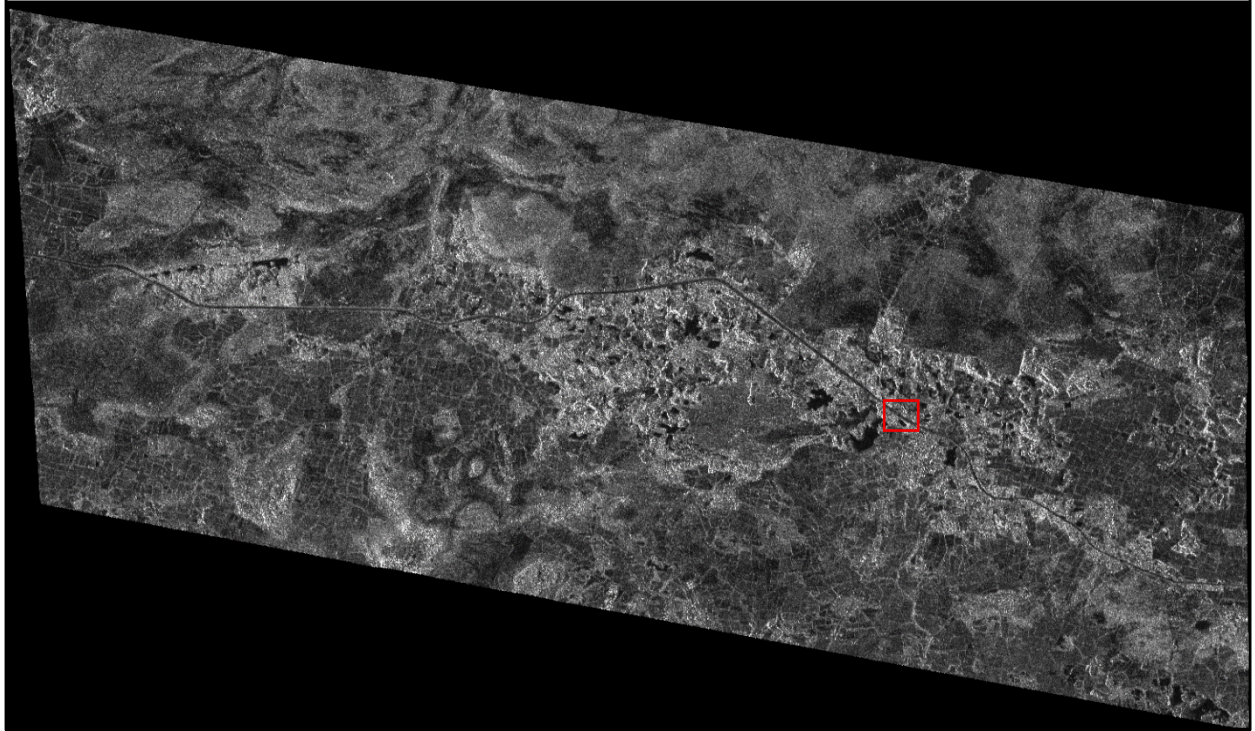
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

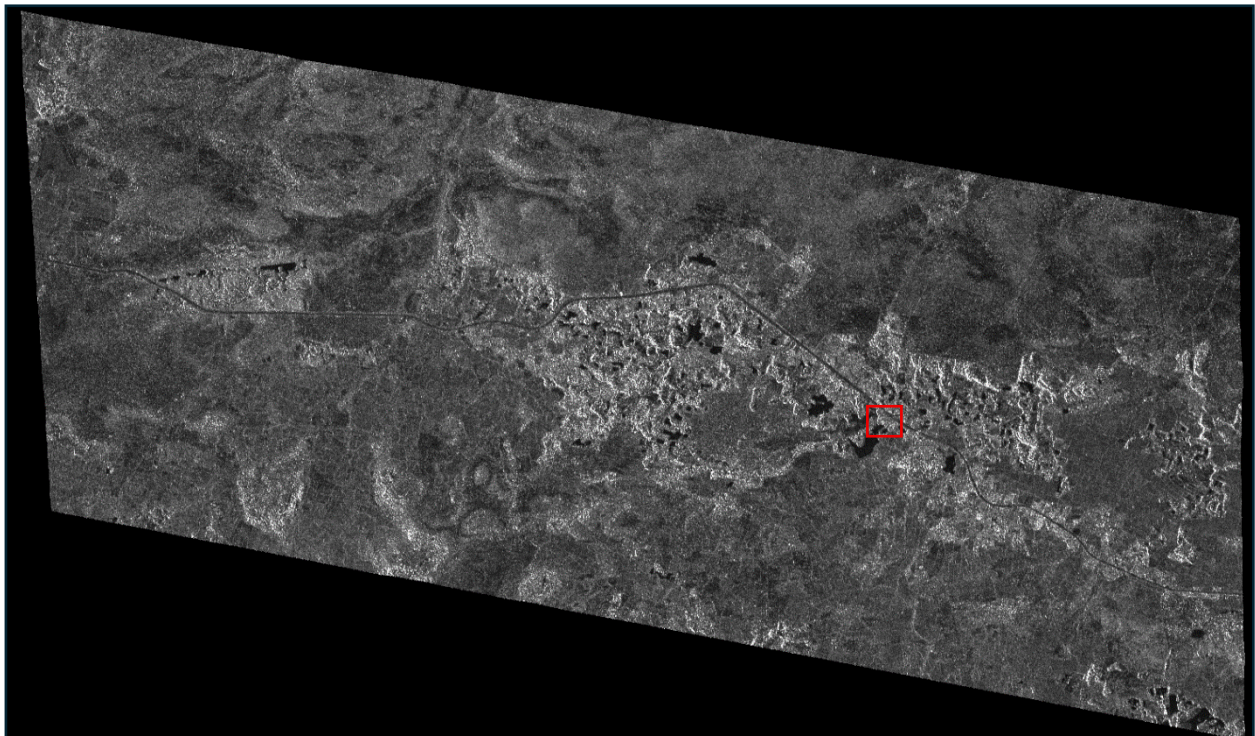
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

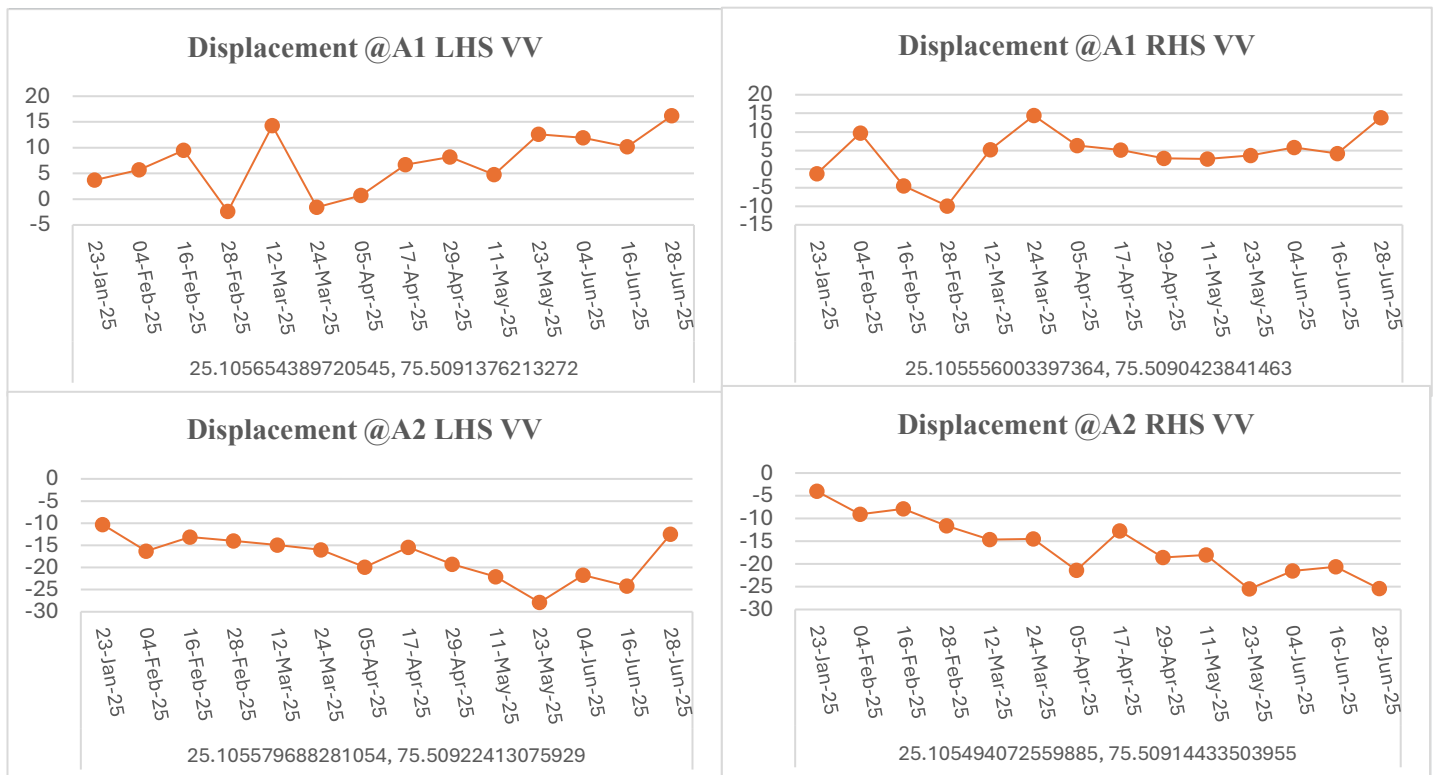
MNB – CH-1022+850

Length – 26.83M

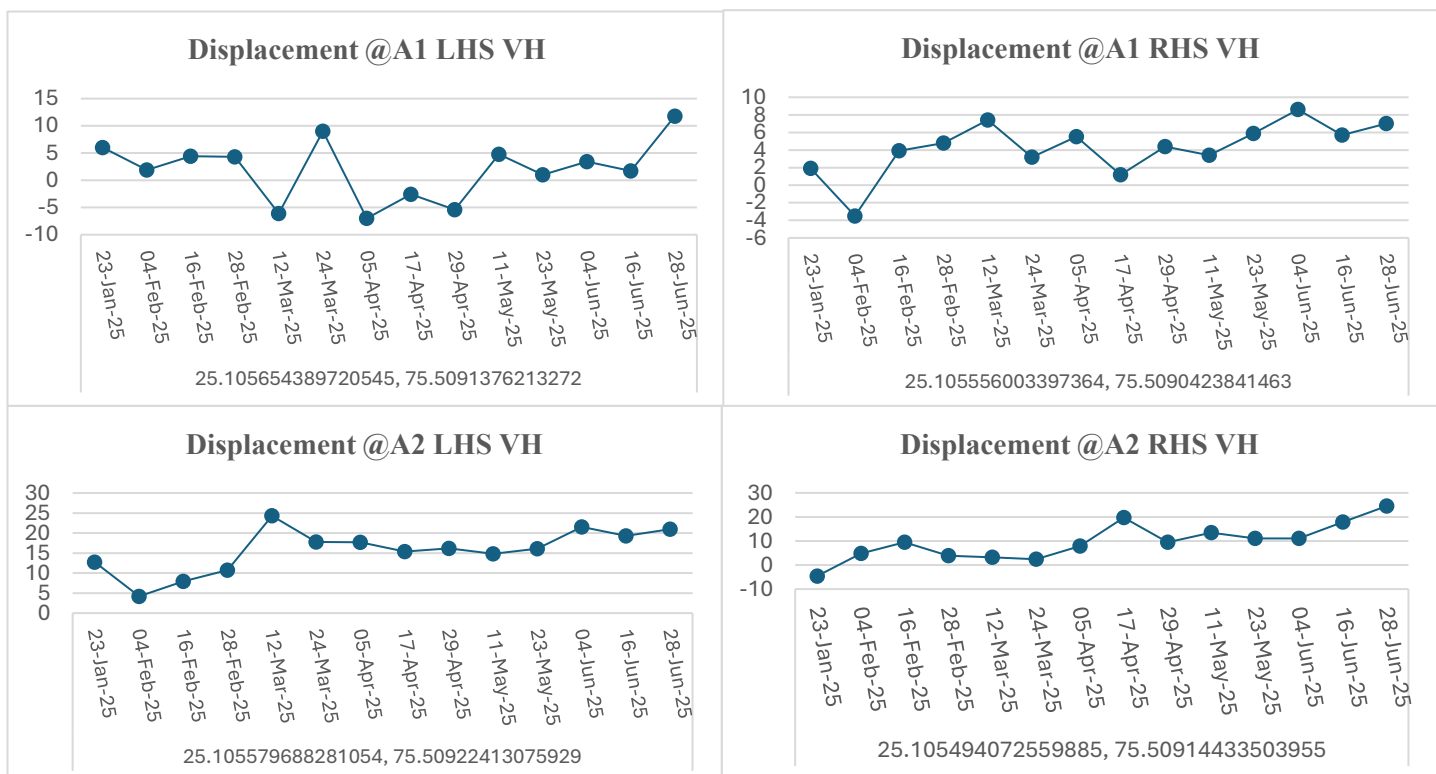


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

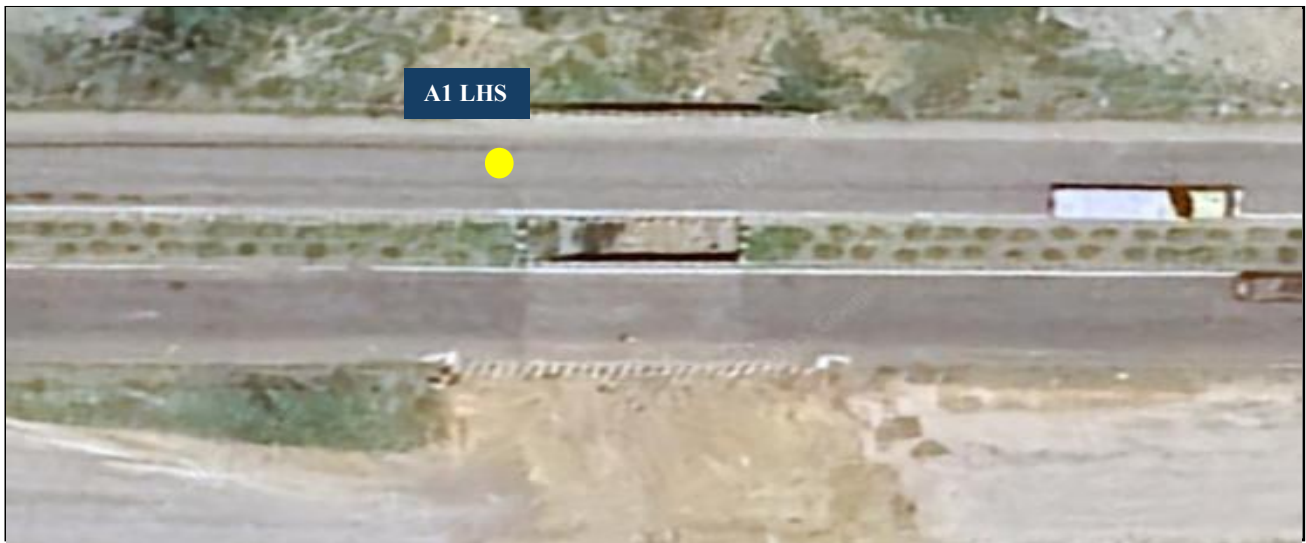
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



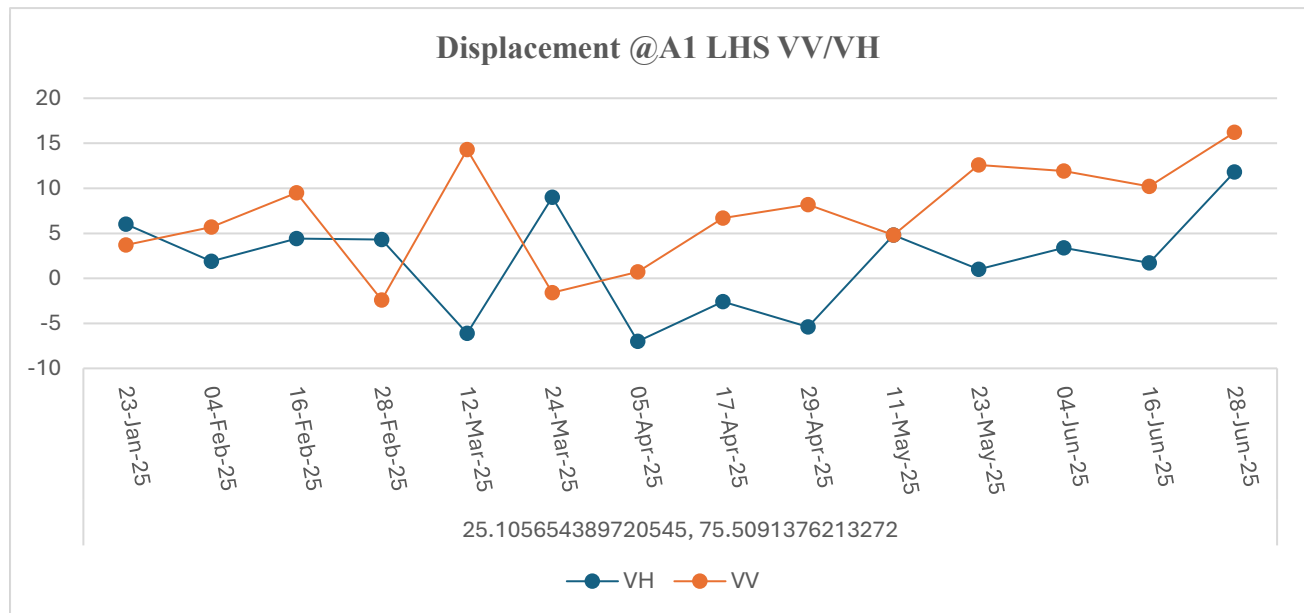
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.105654389720545, 75.5091376213272)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** values range approximately between -2 mm and +16 mm. After minor fluctuations in January and early February, VV increases noticeably in late February, followed by short-term reductions in early March.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits alternating positive and negative behaviour, ranging approximately between -7 mm and +11 mm. The pattern is oscillatory, with periods of recovery and decline.

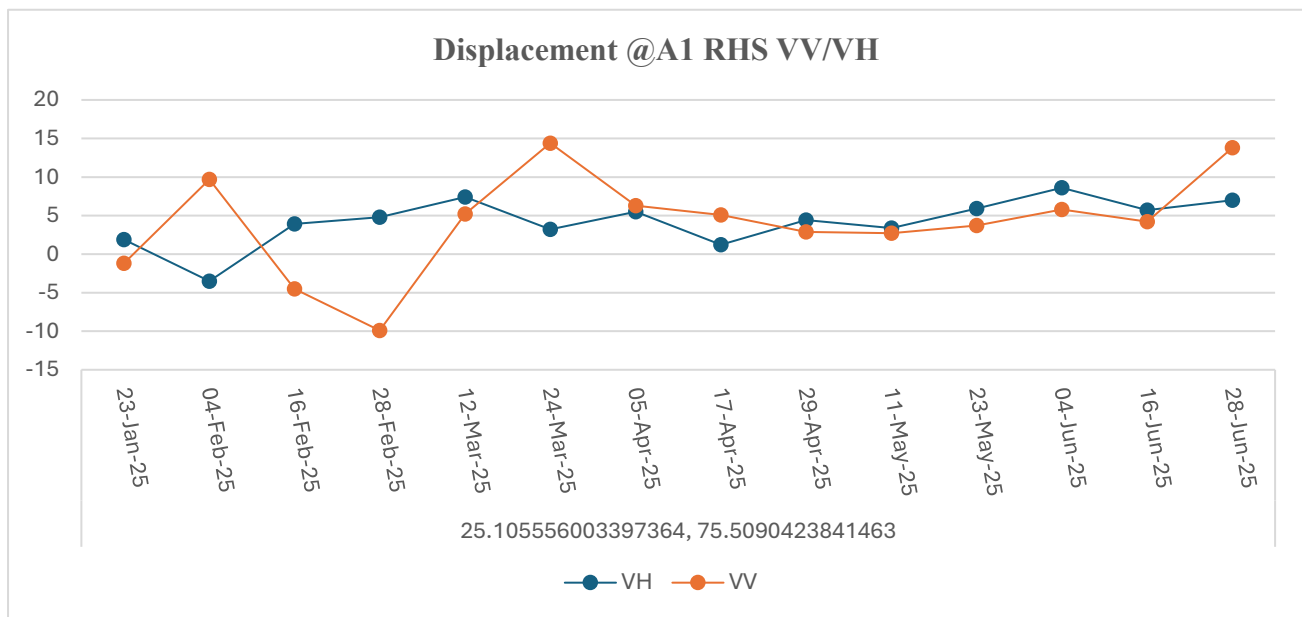
Critical Events & Anomalies:

- **February-March 2025:** VV reaches a local peak (+14 mm) while VH drops sharply to approximately -6 mm, indicating a short-term vertical uplift combined with lateral reversal.
- **June 2025:** Late June Both VV and VH rise simultaneously (VV +16 mm, VH +11 mm), indicating combined vertical uplift and lateral displacement intensification.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.105556003397364, 75.5090423841463)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains moderate and predominantly positive, varying approximately between –3 mm and +9 mm. After an initial dip in early February, VH increases steadily through March and April, followed by minor oscillations with an overall stable to slightly increasing trend toward the end of the monitoring period.
- **Vertical displacement (VV)** VV values range approximately between –10 mm and +14 mm. A pronounced downward movement occurs in late February, followed by a strong recovery in early March.

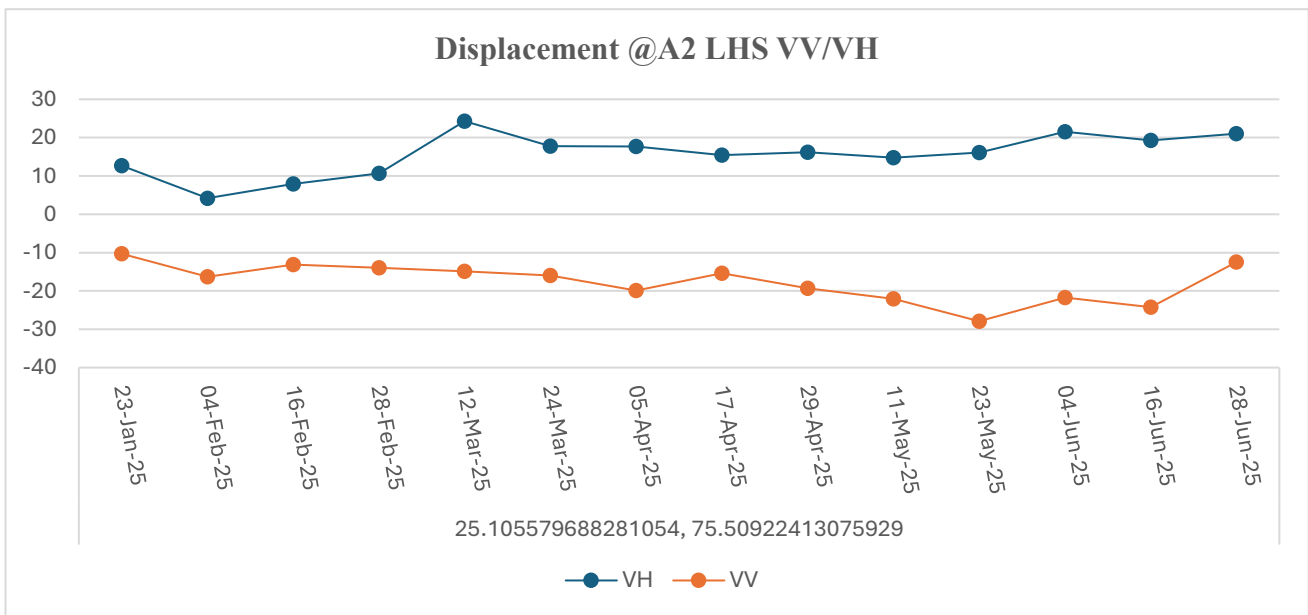
Critical Events & Anomalies:

- **March 2025:** VV experiences a sharp negative excursion (–10 mm) while VH remains moderately positive, indicating a short-term vertical disturbance without significant lateral instability.
- **June 2025:** Both VV and VH show simultaneous increases (VV +13–14 mm, VH +7–8 mm), indicating combined vertical uplift and lateral movement toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.105579688281054, 75.50922413075929)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains consistently positive and relatively stable, ranging approximately between +6 mm and +23 mm.
- **Vertical Displacement (VV)** Vertical displacement shows a consistently negative trend throughout the monitoring period, indicating sustained settlement. VV values range approximately between –10 mm and –30 mm.

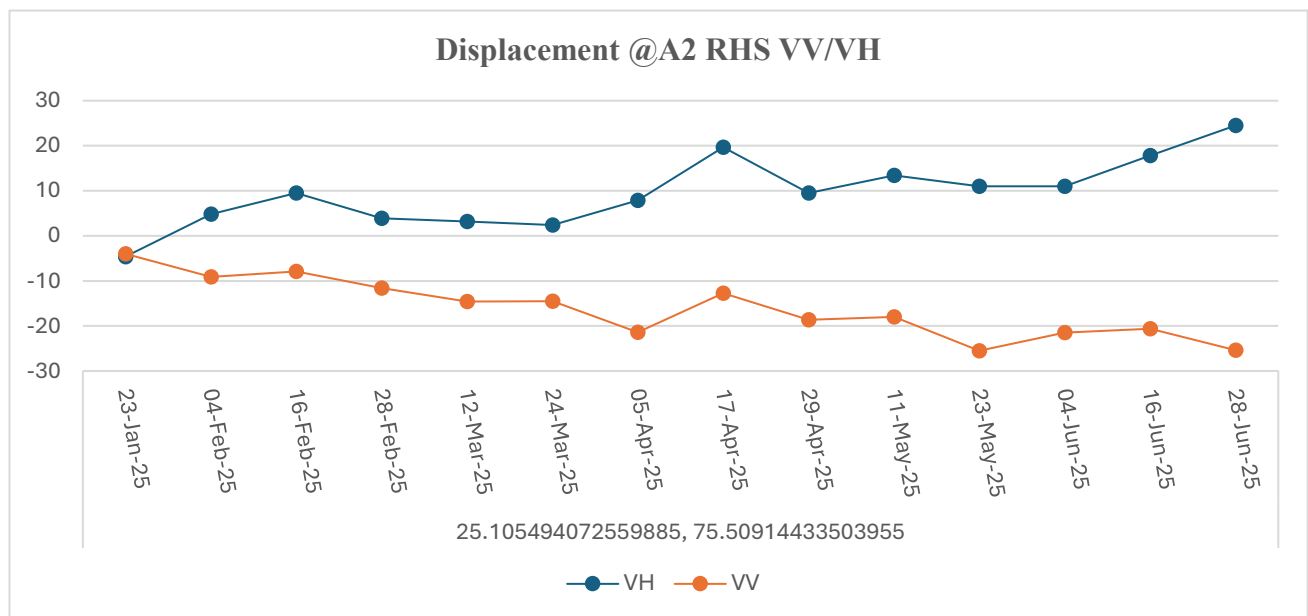
Critical Events & Anomalies:

- **March 2025:** VH reaches an early peak (+24 mm), while VV continues trending downward, highlighting divergent horizontal and vertical behaviour.
- **May 2025:** VV reaches its maximum negative displacement (–30 mm), coinciding with moderately elevated VH values (+15 mm), representing the most critical settlement phase
- **June 2025:** VH attains its highest observed value (+23–24 mm) while VV partially recovers, indicating strong lateral response alongside stabilising vertical movement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.105494072559885, 75.50914433503955)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV values range approximately between –6 mm and –27 mm. A slight recovery is seen toward late June, although values remain significantly negative.
- **Horizontal (VH) displacement** VH shows a general increasing trend from January through June, with intermittent fluctuations.

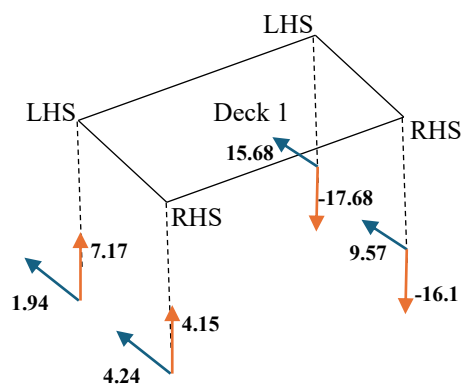
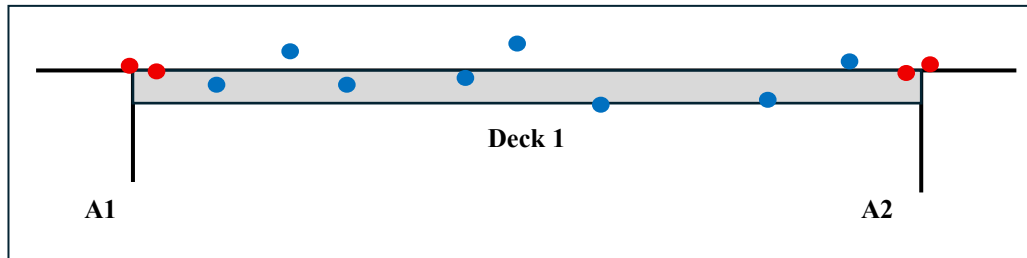
Critical Events & Anomalies:

- **February 2025:** VH increases to around +10 mm, while VV drops below –10 mm, indicating the onset of combined lateral movement and settlement.
- **May 2025:** VV reaches its maximum negative displacement (–27 mm), coinciding with elevated VH values (+12–15 mm), representing the most critical combined displacement phase
- **June 2025:** VH reaches its highest observed value (+24 mm) while VV remains strongly negative, indicating sustained lateral dominance with continued vertical settlement.

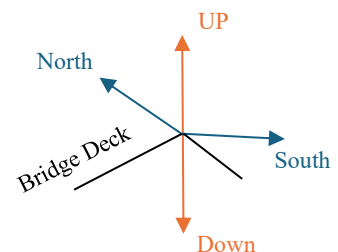
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement across both LHS and RHS regions, indicating differential structural response. The maximum uplift is observed at the upper central deck region, reaching approximately +15.7 mm, indicating pronounced vertical movement at the deck crown. Additional uplift is recorded at the lower LHS region (+7.2 mm) and lower RHS region (+4.2 mm), suggesting moderate vertical response near support zones

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability across the deck width. The LHS region exhibits relatively low lateral movement, ranging between approximately +1.9 mm and +4.2 mm, indicating comparatively stable behaviour.

CONCLUSION:

- The bridge deck slab demonstrates a highly non-uniform deformation pattern, characterised by significant vertical uplift and settlement occurring simultaneously across different regions. The RHS exhibits both greater vertical settlement and increased lateral displacement, indicating a zone of higher structural demand or reduced stiffness